

LUZALEN MARCOS

PhD Candidate in Electrical and Computer Engineering
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PROFESSIONAL SUMMARY

PhD student in Electrical and Computer Engineering with research interests in Artificial Intelligence, Natural Language Processing, Computer Vision, Data Science, and Human-Computer Interaction. Experienced in diverse team collaboration, machine learning model development, and academic teaching.

EDUCATION

PhD Electrical and Computer Engineering

Toronto Metropolitan University, Toronto, ON

Present

Research focus: Artificial Intelligence, Natural Language Processing, Computer Vision, Data Science, Human-Computer Interaction

MASc Electrical and Computer Engineering

Toronto Metropolitan University, Toronto, ON

2022

Thesis: Green Roof Detection Through Satellite Imagery using Faster R-CNN

Bachelor of Computer Engineering minor in Politics

Toronto Metropolitan University, Toronto, ON

Completed

RESEARCH EXPERIENCE

Research Intern

Toronto Metropolitan University, Toronto, ON

Previous

- Assisted in designing data preprocessing machine learning models and algorithms that monitor vital signs of users and alert users for any issues
- Experienced collaborative learning and work with students from different fields within and from students from other universities

TEACHING EXPERIENCE

Teaching Graduate Student

Toronto Metropolitan University, Toronto, ON

Previous

- Psychophysics: Used different teaching tools such as Kahoots!, Padlet to introduce lab concepts to help students increase student engagement during online labs
- Human-Computer Interaction: Presented UI/UX concepts, such as Model-Based Systems Engineering, qualitative and quantitative design analysis and wireframing tools such as Figma and Adobe XD, in a class of at least 20 students
- Digital Systems: Assisted students to increase understanding of concepts of Logic gates, and applied the concepts and FPGA Programming using VHDL. Evaluated lab demonstrations and reports of around 50 students
- Rehabilitation Engineering: Facilitated online labs that assessed students' understanding of rehabilitation concepts. Utilized online tools to increase student engagement and promote active learning
- Database Management Systems: Facilitated offline and online labs that assessed students' understanding by applying concepts learned such as ERD, SQLite3 and XML through labs
- Object-Oriented Engineering Analysis and Design: Assessed students' ability to apply course concepts through lab activities. Clarified lab requirements

PUBLICATIONS AND PRESENTATIONS

Research Projects

- Emotion Classification Through Speech Data Analysis: Explored sound characteristics to determine other features that can be used for Speech Emotion Recognition (SER). Compared various machine learning models to determine the baseline model for speech emotion classification
- Green Roof Detection Through Satellite Imagery using Faster R-CNN: Developed and trained a deep learning model to detect green roofs from satellite imagery for urban sustainability monitoring
- Human-Computer Interaction: Design Process and Ideation: Course project focusing on user-centered design methodologies and prototyping techniques

SKILLS

Programming Languages: SQL, Python, Java, VHDL, C++, JavaScript

Design Tools: Adobe Creative Suite, Figma, Sketch

Specialized: MATLAB, FPGA Development, LabVIEW, TensorFlow, PyTorch

CERTIFICATIONS

- NASA Human Spaceflight Program - National Aeronautics and Space Administration, 2023
- Machine Learning Specialization - DeepLearning.AI, 2023
- Deep Learning Specialization - DeepLearning.AI, 2023
- Python for Everybody Specialization - University of Michigan, 2023
- Data Science Professional Certificate - IBM, 2023
- Google UX Design Professional Certificate - Google, 2023
- Meta Front-End Developer Professional Certificate - Meta, 2023
- Microsoft Azure Fundamentals - Microsoft, 2023
- AWS Cloud Practitioner - Amazon Web Services, 2023
- Project Management Professional - Project Management Institute, 2023
- Agile Project Management - LinkedIn Learning, 2023
- Communication Skills for Engineers - LinkedIn Learning, 2023

LEADERSHIP & VOLUNTEER EXPERIENCE

- Co-President - Engineers Without Borders, Toronto Metropolitan University, 2022-2023
- FEAS Ambassador - Faculty of Engineering and Architectural Science, 2021-2023
- Vice President Academic - Computer Engineering Society, 2021-2022
- Vice President Finance - Computer Engineering Society, 2020-2021
- Senior Computer Officer - IEEE TMU Student Branch, 2021-2022
- Computer Director - Toronto Metropolitan University, 2022-2023
- National Girls Learning Code Mentor - Girls Learning Code, 2021-2023
- Teens Learning Code Mentor - Girls Learning Code, 2021-2023

COMMUNITY INVOLVEMENT

Student Volunteer

GoCode Girl Event

- Participated in planning GoCode Girl, an event that teaches younger students programming by giving ideas on what activities to do to make them understand the logic used in programming
- Guided Girl Guides of Ontario to obtain their Science and Technology: Engineering badges by giving them an insight about engineering and making them assemble circuits using snap circuits

Mentoring*First-Year Student Mentorship*

- Mentored five first-year students or mentees in transitioning from a high school to a university environment by giving them a university tour, links to resources such as clubs and contacts, and tips on where and how to study
- Attended events with mentees that will help them get to know Ryerson, job opportunities and with their studies

Sustainability Initiatives*Enviro-Innovation Case Competition*

- Collaborated with Regenesi TMU, a student-empowered environmental organization, to organize the Enviro-Innovation Case Competition that was participated by more than 10 students